

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2126

Commonwealth Chesapeake 345 kV
August 02, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2126. The Interconnection Customer proposes a 475.0 MW solar photovoltaic facility interconnecting at the Commonwealth Chesapeake 345 kV substation in the Virginia area. The request was received on June 30, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 697 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,106,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: ORANGE BULK / WHITE BLUFF 138kV Line
Contingency: CADY / EASTON 2 345kV
Loading: 118.2% (Rating: 418 MVA)
DFAX: 10.1%
Pre-project loading: 106.7%
2. Facility: SUTHERLAND SPRINGS / STAVE FALLS 138kV
Contingency: BOND-ELK GROVE FLORIN 345/138kV Transformer Bank
Loading: 112.4% (Rating: 386 MVA)
DFAX: 7.4%
Pre-project loading: 103.2%
3. Facility: C W TIPPY / STAVE FALLS 138kV L
Contingency: BOND-ELK GROVE FLORIN 345/138 kV Transformer Bank
Loading: 111.7% (Rating: 396 MVA)
DFAX: 7.4%
Pre-project loading: 102.9%
4. Facility: MCKEE RUN / WHEATLAND STA 138kV
Contingency: CADY / DODD ROAD 345kV
Loading: 111.1% (Rating: 415 MVA)
DFAX: 6.8%
Pre-project loading: 103.2%
5. Facility: NINEMILE POINT / N LEBANON 138kV #1
Contingency: NORTH LEBANON / NINEMILE PT 138kV
Loading: 108.5% (Rating: 685 MVA)
DFAX: 8.7%
Pre-project loading: 102.5%

6. Facility: NINEMILE PT / NORTH LEBANON 138kV #2
Contingency: NINEMILE POINT SUBSTATION / N LEBANON 138kV Cir. 1
Loading: 108.5% (Rating: 685 MVA)
DFAX: 8.7%
Pre-project loading: 102.5%
7. Facility: WHITMORE / EDGEWATER STATION 138kV
Contingency: BOND-ELK GROVE FLORIN 345/138kV XFMR
Loading: 108.3% (Rating: 428 MVA)
DFAX: 7.4%
Pre-project loading: 100.1%
8. Facility: BOND-ELK GROVE FLORIN / FAIRONT 138kV Circuit 1
Contingency: BOND-ELK GROVE FLORIN / FAIRMONT 138kV Line 2
Loading: 108.2% (Rating: 647 MVA)
DFAX: 8.6%
Pre-project loading: 101.9%
9. Facility: BOND-ELK GROVE FLORIN / FAIRMONT 138kV Circuit 2
Contingency: BOND-ELK GROVE FLORIN / FAIRMONT 138kV
Loading: 108.2% (Rating: 647 MVA)
DFAX: 8.6%
Pre-project loading: 101.9%
10. Facility: N LEBANON / FAIRMONT 138kV #1
Contingency: FAIRMONT / N LEBANON 138kV Cir. 2
Loading: 107.9% (Rating: 654 MVA)
DFAX: 8.5%
Pre-project loading: 101.7%
11. Facility: FAIRMONT / N LEBANON 138kV #2
Contingency: FAIRMONT / N LEBANON 138kV Ckt 1
Loading: 107.9% (Rating: 654 MVA)
DFAX: 8.5%
Pre-project loading: 101.7%
12. Facility: HOPE CRK 345/138kV Autotransformer
Contingency: BOND-ELK GROVE FLORIN 345/138kV Autotransformer
Loading: 104.5% (Rating: 404 MVA)
DFAX: 7.4%
Pre-project loading: 95.8%
13. Facility: BOND-ELK GROVE FLORIN 345-138 kV TX
Contingency: CADY / DODD RD 345kV
Loading: 103.1% (Rating: 1,337 MVA)
DFAX: 34.5%
Pre-project loading: 90.8%
14. Facility: WICK DRIVE / CEDARSAUK 138kV line
Contingency: CADY / DODD ROAD 345kV
Loading: 101.4% (Rating: 231 MVA)
DFAX: 6.6%
Pre-project loading: 87.7%
15. Facility: DONNER / NINEMILE PT 138kV
Contingency: DODD ROAD / CADY 345kV Line
Loading: 101.3% (Rating: 615 MVA)
DFAX: 7.4%
Pre-project loading: 95.6%
16. Facility: HOPE CRK / SUTHERLAND SPRINGS 138kV
Contingency: BOND-ELK GROVE FLORIN 345/138 kV TX

Loading: 101.1% (Rating: 418 MVA)

DFAX: 7.4%

Pre-project loading: 92.7%

17. Facility: EDGEWATER STA / C W TIPPY 138kV line

Contingency: BOND-ELK GROVE FLORIN 345/138 kV Transformer

Loading: 100.2% (Rating: 452 MVA)

DFAX: 7.4%

Pre-project loading: 92.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ORANGE BULK / WHITE BLUFF 138kV Line	Rebuild line with higher-capacity conductor	\$16,896,000	9% (\$1,594,000)
SUTHERLAND SPRINGS / STAVE FALLS 138kV	Rebuild line with higher-capacity conductor	\$2,941,000	8% (\$219,000)
C W TIPPY / STAVE FALLS 138kV	Rebuild line with higher-capacity conductor	\$2,473,000	8% (\$184,000)
MCKEE RUN SUBSTATION / WEATLAND STA 138kV	Rebuild line with higher-capacity conductor	\$3,387,000	10% (\$347,000)
NINEMILE POINT / N LEBANON 138kV #1	Reconductor line	\$725,000	6% (\$46,000)
NINEMILE PT / NORTH LEBANON 138kV #2	Reconductor line	\$507,000	6% (\$32,000)
WHITMORE / EDGEWATER STATION 138kV	Reconductor line	\$2,579,000	8% (\$192,000)
BOND-ELK GROVE FLORIN / FAIRMONT 138kV Ckt 1	Reconductor line	\$621,000	6% (\$39,000)
BOND-ELK GROVE FLORIN / FAIRMONT SUBSTATION 138kV Cir. 2	Reconductor line	\$941,000	6% (\$59,000)
NORTH LEBANON / FAIRMONT 138kV Circuit 1	Reconductor line	\$1,005,000	6% (\$63,000)
FAIRMONT / N LEBANON 138kV #2	Reconductor line	\$772,000	6% (\$49,000)
HOPE CRK 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$11,197,000	8% (\$834,000)
BOND-ELK GROVE FLORIN 345-138 kV TX	Replace transformer with higher-rated unit	\$20,509,000	10% (\$1,955,000)
WICK DRIVE / CEDARSAUK 138kV Line	Reconductor line	\$651,000	13% (\$83,000)
NINEMILE PT / DONNER 138kV line	Reconductor line	\$1,257,000	5% (\$68,000)
SUTHERLAND SPRINGS / HOPE CRK 138kV	Reconductor line	\$3,060,000	8% (\$228,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
EDGEWATER STA / C W TIPPY 138kV line	Reconductor line	\$1,528,000	8% (\$114,000)
	Total allocated to project		\$6,106,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.